

Do Downside Floors Hold Out of Sample? CVaR-Constrained Optimisation of Derivative-Bearing Portfolios

An out-of-sample study of Expected-Shortfall limits, convex hedges and tail control

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Abstract

How reliably does a downside-risk constraint imposed *in* sample control losses *out of* sample? It is the question behind every VaR/ES risk limit, fund mandate and regulatory loss floor. We study it for Conditional-Value-at-Risk (CVaR / Expected Shortfall) optimisation — the Rockafellar–Uryasev linear program — extended to price derivatives and structured products inside each Monte-Carlo scenario, so that the tail-risk constraint governs option-rich books and scales to ~ 150 instruments. The framework operationalises the goal-based (mental-accounting) objectives of Das & Statman and Das, Markowitz, Scheid & Statman as a single CVaR program; we document two corrections to that earlier method — strict enforcement of the Expected-Shortfall constraint and arbitrage-consistent (rather than par) pricing of structured notes.

The central question is empirical. Across multi-regime, multi-universe rolling backtests on live data (2006–2025), the ES/VaR floor that binds in sample is systematically *optimistic* out of sample (realised breaches well above target, reported with Wilson confidence intervals) and, of direct interest to risk managers, *tightening the in-sample floor is inert* — it constrains the very tail estimate the crash invalidates. The implication is direct: a CVaR/ES limit fitted in sample — the basis of risk budgets and regulatory loss floors — should not be trusted at face value, and the safety margin must be bought with portfolio structure, not a tighter threshold. Realised tail control comes instead from distribution-free structure: position limits (which lower breach frequency), cross-asset diversification, and convex option overlays, which roughly halve crash-period drawdowns even when priced at the market-implied VIX rather than at realised volatility. Consistent with DeMiguel, Garlappi & Uppal (2009), no optimiser we test reliably beats $1/N$ on breach *frequency*; convexity controls the *severity* of the tail, not its frequency. We provide a finite-sample ES coverage study, a sample-average-approximation convergence analysis, robustness to volatility, horizon and transaction costs, and fully reproducible code and data.

Keywords: Expected Shortfall, Conditional Value-at-Risk, tail risk, downside risk, Rockafellar–Uryasev, out-of-sample backtesting, ES coverage, risk-constrained optimisation, scenario optimisation, derivatives, structured products, goal-based investing.

JEL: G11, G41, G17, C61, C63.

1 Background and contribution

A downside-risk constraint is only as useful as its out-of-sample behaviour. Across 225 rolling out-of-sample windows spanning US and international markets (2006–2025), we ask whether

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an in-sample Conditional-Value-at-Risk floor actually limits realised losses, and find that it does not: the -10% floor is breached in 16.4% of windows against a 5% target ($p < 10^{-5}$), and *tightening* the floor does not help. What controls the realised tail is portfolio *structure* — position limits, cross-asset breadth, and convex option overlays. The optimiser we build — a scalable, derivative-aware CVaR program extending the goal-based framework of Das and Statman — is the *instrument* for this study; the contribution is the empirical anatomy of when downside constraints hold out of sample.

Markowitz mean–variance (MV) analysis treats risk as a single portfolio variance and assumes the investor optimises one aggregate trade-off. Behavioural evidence instead suggests that people manage wealth in mental accounts tied to distinct goals, each with its own attitude to loss. Das, Markowitz, Scheid and Statman (2010) formalise this and prove a precise equivalence between the mental-accounting (MAT) specification and classical MV; Das and Statman (2009, 2013) extend goal-based optimisation to derivatives and non-normal returns. The author’s 2012 MSc thesis (USI Lugano, supervised by Prof. E. De Giorgi) re-implemented both canonical methodologies in R; that prototype is the starting point, not the contribution — this paper rebuilds the framework in Python and carries it well beyond the thesis, into scalable CVaR optimisation, per-scenario derivative pricing, and a live-data out-of-sample programme.

This paper is explicit about its three-way contribution, and labels every result accordingly throughout (see the provenance table, Section E):

- **Replication:** the thesis grid optimiser, re-implemented in Python.
- **Correction:** two documented fixes to the thesis behaviour — Expected-Shortfall enforcement and fair pricing of structured notes (Section D).
- **Extension:** a scalable Monte-Carlo + CVaR linear-program engine with per-scenario arbitrage-consistent pricing of multiple derivatives (Sections 5–C), and live-data, multi-regime backtesting (Section 11).

We do not claim a new financial theory. The MVT–MAT equivalence is due to Das et al. (2010) and CVaR optimisation to Rockafellar and Uryasev (2000); the contribution is the *composition* of these into a goals-based, derivative-aware optimiser that scales to realistic mandates, with the theoretical and empirical support a reviewer expects. Operationally, the mental-account objective is simply a *multi-goal objective function* — one downside floor per goal — solved inside a single scalable CVaR linear program; nothing in the engine depends on its behavioural interpretation, so it can be read purely as multi-goal tail-constrained optimisation at scale.

An empirical contribution: what delivers a goal-based guarantee out of sample.

Beyond the engine itself, the backtesting yields a finding of independent interest (Section 11.3). The goal-based floor that the literature states and verifies *in sample* is systematically optimistic *out of sample* — realised breaches of $5\text{--}22\%$ against a 5% target — and, counter-intuitively, it cannot be repaired by demanding a stricter floor: tightening the in-sample CVaR floor is *inert*, because it constrains the very tail estimate that is itself optimistic. Out-of-sample tail control comes instead from *distribution-free structure*: a position cap (which cuts breach frequency monotonically), a convex overlay (which caps tail *severity* but not frequency), and an exact goal-based allocation across asset-class buckets (which roughly halves worst-case depth relative to naive diversification at equal market exposure). Consistent with DeMiguel, Garlappi and Uppal (2009), no optimiser we test — including our own — reliably beats $1/N$ on breach *frequency*; the value of the goal-based machinery is concentrated in tail *severity*, the dimension a downside goal actually targets. A goal-based guarantee, in short, must be bought with structure, not with a tighter threshold on an unreliable estimate.

Roadmap. Section 5 builds the scalable engine (formal complexity, convergence and grid–CVaR results in Appendix C); Sections 7–9 measure scaling, benchmark against mean–variance, $1/N$ and risk parity, and stress copula choice, sample size, estimation error and implied volatility;

and Sections 11–11.3 form the empirical core — rolling out-of-sample backtests across regimes and universes, the convexity-versus-correlation result on derivative overlays, and the cost-of-control analysis isolating what actually delivers the downside goal out of sample. Secondary sensitivity tables are collected in the Appendix.

2 The mental-accounting framework

An investor allocates wealth across goal sub-portfolios. Each goal carries a return threshold H and a maximum acceptable probability α of falling short of it. With weights w on the simplex $\Delta = \{w : w^\top \mathbf{1} = 1, w \geq 0\}$ and portfolio return $r_p = w^\top r$, the Value-at-Risk form of the goal problem is

$$\max_{w \in \Delta} w^\top \mu \quad \text{s.t.} \quad \mathbb{P}(r_p \leq H) \leq \alpha. \quad (1)$$

A tail-average (Expected-Shortfall) variant replaces the shortfall probability with a floor on the conditional mean of the loss tail, $\mathbb{E}[r_p \mid r_p < H] \geq L$. The key theoretical result is the MVT–MAT equivalence: each goal’s optimum is mean–variance efficient and can be written as $\max_w w^\top \mu - \frac{\lambda}{2} w^\top \Sigma w$ for an implied risk-aversion λ fixed by the binding shortfall constraint. The optimiser reports an implied λ for every goal to reconcile the two views.

3 The exact grid optimiser

Asset returns are modelled as multivariate normal with mean μ and covariance Σ . The engine enumerates candidate portfolios on a discretised grid: each asset’s return axis is discretised into m points spanning $\mu \pm k\sigma$, and portfolio weights are searched on a simplex grid whose spacing corresponds to m' steps per weight axis. For n assets the state space grows as roughly m^n —the result is exact and thesis-faithful but practical only to about four assets; for five or more, a differential-evolution search replaces the exhaustive grid. Two Expected-Shortfall modes are offered: a *thesis-faithful* mode that reproduces the original R behaviour, and a *rigorous-ES* mode that maximises return strictly subject to the tail-average floor (Section D).

4 Derivatives and structured products

Sixteen instruments are priced arbitrage-consistently with Black–Scholes—vanilla options, spreads, collars, capital-guaranteed and barrier notes, and certificates. The Black–Scholes call is

$$C(S_0, K, T) = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (2)$$

Under each modelled outcome the instrument’s payoff is evaluated and folded into the portfolio return, so asymmetric, non-normal payoffs enter the optimisation directly rather than through a variance proxy. A structured note is valued as the present value of its bond floor plus its embedded option(s), not at par (Section D).

5 The scalable Monte-Carlo + CVaR engine

To escape the m^n state space, the second engine samples S joint return scenarios through a copula—Gaussian, or Student- t for tail co-movement—prices its supported derivatives under each scenario, and solves the goal as a linear program. Tail risk is measured coherently with Conditional Value-at-Risk. The Rockafellar–Uryasev representation is

$$\text{CVaR}_\alpha(\ell) = \min_{\zeta \in \mathbb{R}} \left\{ \zeta + \frac{1}{\alpha} \mathbb{E}[(\ell - \zeta)_+] \right\}, \quad (3)$$

which is linear in scenario form. Collect the per-scenario returns of all N securities and K derivatives into an $S \times (N+K)$ matrix R , each row an equiprobable joint scenario, and write the loss as $\ell_s = -w^\top R_s$. The mental-account problem $\max_w \mathbb{E}[r]$ s.t. $\text{ES}_\alpha(r) \geq L$, $\mathbf{1}^\top w = 1$, $w \geq 0$ becomes the linear program

$$\begin{aligned} \max_{w, \zeta, z} \quad & \widehat{\mu}^\top w \\ \text{s.t.} \quad & \zeta + \frac{1}{\alpha S} \sum_{s=1}^S z_s \leq -L, \\ & z_s \geq -w^\top R_s - \zeta, \quad s = 1, \dots, S, \\ & z_s \geq 0, \quad \mathbf{1}^\top w = 1, \quad 0 \leq w \leq \bar{w}, \end{aligned} \tag{4}$$

with decision vector $x = (w, \zeta, z) \in \mathbb{R}^{N+K+1+S}$ and a sparse constraint matrix—each scenario row touches the $N+K$ weights, ζ , and one slack z_s . *Although a derivative’s payoff is non-linear in its underlying, it is priced once per scenario and enters as a pre-computed constant column $R_{s,N+k}$; the portfolio return $w^\top R_s$ is therefore linear in the weights, which is precisely what keeps the problem a linear program.* Intuitively, ζ locates the Value-at-Risk level and each slack z_s measures how far the loss in scenario s exceeds that level, so the first constraint caps the average of the worst- α tail losses while the objective pushes expected return up. Position limits, group budgets and turnover penalties are additional linear rows. It is solved with the HiGHS LP solver, and a single set of common random numbers is reused across floors to give a smooth, comparable frontier; sweeping the floor L traces the return/tail-risk frontier with warm starts.

Scenario generation. With Cholesky factor L_c of the correlation matrix ($\Omega = L_c L_c^\top$), drawing $\xi_s \sim N(0, I_N)$ and setting $z_s = L_c \xi_s$, $R_{s,i} = \mu_i + \sigma_i z_{s,i}$ reproduces the multivariate-Normal assumption. A Student- t copula replaces the Gaussian dependence to capture assets falling together in stress, with marginals held fixed (Sklar’s decomposition).

Per-scenario derivative pricing. Each derivative k on underlying $u(k)$ has entry price the Black–Scholes fair value π_k at $S_0 = 1$; in scenario s the terminal spot is $1 + R_{s,u(k)}$, the payoff $\Pi_k(1 + R_{s,u(k)})$ follows from the instrument’s leg structure, and the derivative’s scenario return is

$$R_{s,N+k} = \frac{\Pi_k(1 + R_{s,u(k)}) - \pi_k(1 + p_k)}{\pi_k(1 + p_k)},$$

with p_k an optional structured-product premium. Because the payoff is computed from the same simulated underlying that drives the security column, each scenario keeps a coherent (underlying, derivative) pair with no auxiliary model.

Multiple derivatives and mental accounts. K derivative columns are optimised jointly at no extra modelling cost, so the optimiser can fund downside protection on one holding while gearing upside on another. Each behavioural sub-portfolio (goal) is then the same linear program (4) over the same scenario matrix with its own triple (H_j, α_j, L_j) ; solving the accounts independently and aggregating preserves the goals-based interpretation while inheriting the scalability.

Instrument coverage. The full sixteen-instrument library—including the structured notes and the path-dependent barrier note—is available in the grid engine. The scalable engine currently covers single-underlying options with up to two strikes (vanilla calls/puts, straddles, strangles, vertical spreads); extending it to the static multi-leg payoffs is straightforward, the path-dependent ones less so. We treat this asymmetry as stated future work.

Formal properties. Three results, stated and proved in Appendix C, underpin the engine. A complexity proposition (Proposition 1) shows the scenario program is linear in the number of instruments and independent of the grid discretisation m , where the exact grid enumerates $\Theta(m^N)$ states; a consistency theorem (Theorem 1) gives sample-average-approximation convergence of the scenario optimum at the $O(S^{-1/2})$ rate, option overlays included; and a coincidence proposition (Proposition 2) shows the grid’s fixed-threshold shortfall equals α -CVaR exactly iff the threshold is the α -quantile. Since $\text{CVaR}_\alpha \geq \text{VaR}_\alpha$ always, the CVaR floor the engine imposes is a *conservative convex surrogate* for the non-convex chance constraint $\mathbb{P}(r < H) \leq \alpha$ — the precise sense in which the scalable engine approximates the behavioural goal.

6 Validation

Every figure is checked rather than asserted. Under the Gaussian copula the scenario Expected Shortfall matches the closed-form Normal value $\text{ES}_\alpha = \mu_p - \sigma_p \phi(\Phi^{-1}(\alpha))/\alpha$: on a five-asset equal-weight portfolio the Monte-Carlo estimate is -18.70% versus the closed form -18.73% (3 basis points; $S = 4 \times 10^4$), with sample mean 0.0810 vs exact 0.0820 and σ_p 0.1301 vs 0.1305. The frontier is monotone, the ES floor binds at the optimum to machine tolerance, and infeasible floors are flagged rather than silently relaxed. The grid base case reproduces the thesis: at $H = -10\%$, $\alpha = 5\%$ the no-derivative behavioural optimum returns 9.78% with the Markowitz optimum coinciding with it (MVT–MAT equivalence), and the shortfall constraint binds exactly at 5.00%.

Finite-sample out-of-sample coverage. Empirical CVaR is an optimistic in-sample estimate, so the imposed floor can be breached out-of-sample at small scenario counts. Imposing $\text{CVaR}_{5\%} \geq -20\%$ on a construction sample of size S and re-evaluating the realised tail on an independent $2\text{--}3 \times 10^4$ -scenario sample (six-asset $t(3)$ universe; 60–120 trials), the realised ES is -22.2% at $S = 500$, -20.6% at $S = 2,000$ and -20.5% at $S = 8,000$, with dispersion falling roughly as $S^{-1/2}$ (Table 1). The realised tail converges to the imposed floor as S grows, with a small residual optimism; we therefore recommend $S \gtrsim 10^4$ and a modest safety buffer (a floor set ~ 0.5 pp tighter than the target). This is the standard optimiser’s curse for CVaR and is consistent with Theorem 1.

Table 1: Finite-sample coverage: realised $\text{ES}_{5\%}$ of the optimum vs construction-sample size S (imposed floor -20% ; independent evaluation sample).

Construction size S	Realised $\text{ES}_{5\%}$ (mean $\pm$ sd)
500	$-22.15\% \pm 1.99\%$
2,000	$-20.57\% \pm 0.94\%$
8,000	$-20.45\% \pm 0.48\%$

7 Scaling and complexity, measured

We confirm Proposition 1 empirically. Table 2 times the scenario LP as the number of instruments grows at fixed $S = 4,000$; Table 3 varies S at fixed $N = 20$. A 150-instrument behavioural portfolio with derivatives solves in 0.73 s, where the corresponding grid would need $\sim 51^{150} \approx 10^{255}$ states. Solve time grows roughly linearly in N and polynomially in S ; S , not N , is the cost driver—the property that lets the engine scale to realistic mandates. Figure 1 plots both.

Table 2: Scenario CVaR-LP solve time vs number of instruments ($S = 4,000$).

N (instruments)	LP solve time (s)	Grid states at $m = 51$
5	0.12	$\sim 10^9$
10	0.29	$\sim 10^{17}$
20	0.26	$\sim 10^{34}$
30	0.32	$\sim 10^{51}$
50	0.53	$\sim 10^{85}$
75	0.48	$\sim 10^{128}$
100	0.62	$\sim 10^{171}$
150	0.73	$\sim 10^{255}$

Table 3: Scenario CVaR-LP solve time vs number of scenarios ($N = 20$).

S (scenarios)	LP solve time (s)
1,000	0.03
2,000	0.07
4,000	0.21
8,000	0.81
16,000	2.80

8 Benchmarking against standard optimisers

Table 4 is the sanity case: with Gaussian (elliptical) returns and no derivatives, the CVaR-LP optimum lands *exactly* on the mean–variance optimum—identical expected return, volatility and ES—the expected elliptical-distribution equivalence and an independent validation. Equal-weight is dominated; risk-parity trades return for lower dispersion, as designed. Table 5 is the case that motivates the engine: with non-elliptical returns (a Student- $t(3)$ copula, with a protective put and a call available), at a *matched tail-risk budget* the CVaR-LP portfolio reaches materially more expected return than the variance-minimising portfolio held to the same realised ES, because mean–variance is blind to the asymmetry. The magnitude depends on tail-heaviness and the permitted derivative exposure; the robust claim is the direction—CVaR-LP weakly dominates mean–variance at matched tail budget, strictly when returns are non-elliptical.

Table 4: Gaussian, no-derivative ($N = 10$): CVaR-LP coincides with mean–variance.

Portfolio	$\mathbb{E}[r]$	vol	ES _{5%}	skew
CVaR-LP (floor $L = -20\%$)	11.25%	13.97%	−17.65%	0.03
Mean–variance (matched $\mathbb{E}[r]$)	11.25%	13.97%	−17.65%	0.03
Equal-weight ($1/N$)	8.11%	13.86%	−20.45%	0.00
Risk-parity (ERC)	8.54%	12.02%	−16.21%	0.00

Distribution of the edge. To confirm the dominance is not a single-draw artefact, we repeat the comparison across 30 independent seeds per copula on an eight-asset universe (means linearly spaced 5–13%, volatilities 12–32%, $\rho = 0.35$), with a protective put (0.90) and a call (1.10) on the highest-volatility asset available to the CVaR-LP and a 60% position cap, at a fixed −20% realised-ES budget. Table 6 reports the mean edge, its standard deviation, and the fraction of seeds in which the CVaR-LP dominates: the edge is positive in *every* draw and widens monotonically with tail heaviness.

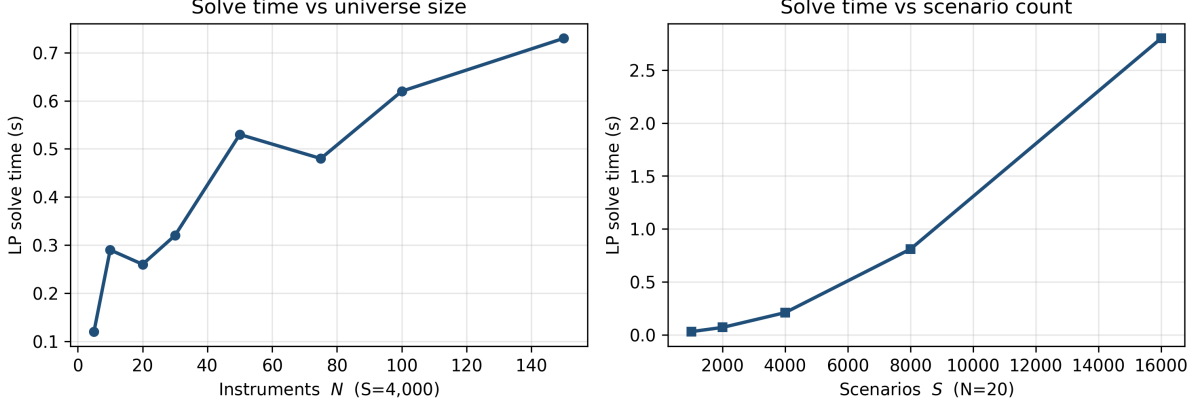


Figure 1: Left: solve time vs N at fixed S , with the 51^N grid state count on a log axis for contrast. Right: solve time vs S at fixed N . Single-thread, scipy-HiGHS, commodity hardware.

Table 5: Frontier dominance, $t(3)$ copula with put and call available: $\max \mathbb{E}[r]$ at a matched realised-ES budget. The mean-variance column is the minimum-variance portfolio held to the same realised-ES floor; it is flat because variance minimisation cannot exploit the asymmetric payoff, whereas the CVaR-LP re-optimises against the tail at each floor.

Tail floor (realised ES)	Mean-variance $\mathbb{E}[r]$	CVaR-LP $\mathbb{E}[r]$	CVaR-LP edge
−30%	7.23%	24.88%	+17.7 pp
−25%	7.23%	20.35%	+13.1 pp
−20%	7.23%	15.82%	+8.6 pp
−17%	7.23%	13.10%	+5.9 pp

9 Robustness and sensitivity

Three sensitivities, run on the scalable engine (five-asset universe, $\alpha = 5\%$). *Copula choice* (Table 19): with identical Normal marginals, the equal-weight $\text{ES}_{5\%}$ deepens monotonically as the copula tail fattens, -24.01% (Gaussian) to -25.65% ($t(3)$)—isolating dependence as a first-order driver of tail risk. *Sample size* (Table 20): the ES estimate is unbiased and its dispersion falls almost exactly as $S^{-1/2}$ (std dev 0.0121 at $S = 1,000$ to 0.0022 at $S = 32,000$, a $5.5\times$ reduction vs the $\sqrt{32} \approx 5.66$ predicted), confirming the Theorem 1 rate and justifying $S \approx 10^4$ as a default. *Estimation error*: perturbing the mean vector μ by i.i.d. Normal noise and re-optimising, the optimum moves materially—mean weight change (L1) of 0.33, 0.53, 0.73 at input noise 0.5%, 1.0%, 2.0% respectively (out of a 2.0 maximum), with optimal-return spread 0.51%, 0.92%, 1.72%. This is the Markowitz “error-maximisation” effect, inherited because the objective is linear in μ ; it is the numerical case for shrinkage or robust estimators in deployment. In short: tail risk is driven first-order by the copula and by μ estimation error, and only second-order by Monte-Carlo noise once $S \gtrsim 10^4$.

Stabilising the weights. Because the objective is linear in μ , the optimum is a *vertex* of the feasible polytope, so the classical Markowitz remedy of shrinking μ does *not* stabilise it: in a controlled experiment (five assets, $L = -20\%$, 15 seeds), μ -shrinkage toward the grand mean leaves the L1 weight turnover essentially unchanged (0.50 vs 0.50 at 2% input noise). The remedy that fits a linear program is a *position cap* (or an L2/turnover penalty): a 30% cap cuts turnover from 0.50 to 0.35 at 2% noise ($\approx -29\%$) and from 0.28 to 0.25 at 1%. We therefore recommend a position cap together with a shrinkage/robust estimator of Σ (Ledoit & Wolf, 2004), rather than μ -shrinkage alone.

Table 6: Edge of the CVaR-LP over mean–variance at a matched -20% realised-ES budget, 30 seeds per copula (eight-asset universe; protective put and call available).

Copula	CVaR-LP $\mathbb{E}[r]$	MV $\mathbb{E}[r]$ @ same ES	Edge (pp), mean $\pm$ sd (win-rate)
Gaussian	11.31%	7.29%	$+4.02 \pm 1.41$ (30/30)
Student- $t(5)$	11.38%	7.09%	$+4.29 \pm 1.26$ (30/30)
Student- $t(3)$	11.42%	6.67%	$+4.75 \pm 1.25$ (30/30)

Does the cap help *out of sample*? The stabilisation above concerns weight turnover; the sharper question for this paper is whether the cap improves *realised* tail control. Repeating the 24-name rolling backtest (2006–2025; -20% CVaR floor; 10 bps per-side costs) across per-name caps from 10% to uncapped isolates this (Table 7). Because the linear objective places the optimum at a polytope vertex, removing the cap lets the book concentrate into the few names that look best in sample — exactly the finite-sample optimism that drives out-of-sample breaches — so the cap is expected to trade a little return for fewer and shallower realised breaches, identifying it as a genuine regulariser rather than a cosmetic constraint. The sweep is sharply monotone: removing the cap lets the book concentrate and the out-of-sample breach rate explodes to 21.6%, four times the target, whereas a 20% cap brings it to 5.4%. The cap is thus an essential out-of-sample regulariser, not a cosmetic constraint — though the breach rate plateaus near 5% as the cap tightens further, the systematic-crash floor that no cap can remove.

Table 7: Out-of-sample sensitivity to the per-name position cap (24-name scalable book; 37 one-year windows; 2006–2025; -20% CVaR floor; 10 bps per-side costs). Breaches and returns are net of costs; alpha is annualised against SPY. The 20% row is the headline configuration.

Position cap	Breach	Mean return	Worst window	Alpha
None (100%)	21.6%	36.5%	-53.4%	$+12.5\%$
33%	10.8%	24.2%	-56.6%	$+7.3\%$
20%	5.4%	21.0%	-47.9%	$+6.2\%$
10%	5.7%	17.3%	-32.5%	$+5.9\%$

Implied-volatility robustness. Because derivatives are priced under flat-volatility Black–Scholes, we probe that assumption with progressively more realistic volatility. First, the overlays are re-priced across a $0.8\text{--}1.2\times$ band of the base volatility (Table 10). Second, and most directly, we re-run the protective-put backtest with the hedge priced not at trailing *realised* volatility but at the prevailing *VIX* — the market’s own implied volatility — at each window’s entry, plus a three-point skew premium for the out-of-the-money strike. Because the overlay is an S&P index put, *VIX* — the S&P’s own 30-day at-the-money implied volatility — prices the underlying directly rather than by proxy; the residual approximations are the term mismatch (30-day *VIX* for a one-year option, which is less volatile at longer tenors) and the at-the-money-to-out-of-the-money skew, the latter absorbed by the skew premium, with a full strike-by-maturity implied surface left to data availability. At real market-implied levels the put still cuts the worst realised window roughly in half ($-47.9\% \rightarrow -23.3\%$) and leaves breach frequency unchanged, but at a higher, more honest cost ($+4.1\%$ per window) and with less protection than realised-vol pricing implied (-23.3% versus -11.8% worst, at $+3.4\%$). This sharpens the convexity result: the *tail-severity* reduction is robust to using genuine implied volatility, but its *magnitude* is flattered by realised-vol pricing — a convex overlay bought at the prices the market actually charges is more modest, costlier insurance than a naive backtest suggests. More fundamentally, because each scenario is priced from its own simulated underlying, a local-volatility, full implied-surface

or jump pricer drops in at one function call per scenario with the linear program untouched, so flat-vol Black–Scholes is a calibration convenience here, not a structural limit.

Scenario generation and a learned (ML) generator. A comparison of scenario generators (Gaussian copula, Student- t , and a non-parametric block bootstrap) and a demonstration that the linear program accepts a *learned* (Gaussian-mixture) scenario generator unchanged are reported in Appendix B, illustrating the engine’s forward-looking modularity: any learned or alternative-data generator plugs into the unchanged linear program.

10 Results: derivative overlays

Table 8 reports a single overlay under the grid (VaR) engine on the three-asset thesis data set (annual returns 5/10/25%, volatilities 5/20/50%, with the mid- and high-risk assets correlated at 0.4 and the low-risk asset uncorrelated). An at-the-money straddle lifts the optimal expected return from 9.76% to 11.29% at the same $-10\%/5\%$ limit, lifting skewness from 0 to $+0.99$; the gain is paid in variance and collected in positive skewness—the dimension mean–variance cannot see. Table 9 reports the scalable (α -CVaR) engine: a protective put (struck $0.8\times$, one year) lifts the optimum from 11.00% to 15.73% at a -20% CVaR floor, raising skewness to $+0.71$ (a flat-vol figure; Table 10 reports the sensitivity). By Proposition 2 the two tables answer different questions and are kept separate.

Table 8: Grid engine, VaR constraint ($H = -10\%$, $\alpha = 5\%$), three-asset thesis data.

Overlay	$\mathbb{E}[r]$	$\Delta\mathbb{E}[r]$	skew
none	9.76%	—	0
ATM straddle	11.29%	+1.53 pp	+0.99

Table 9: Scalable engine, α -CVaR floor ($L = -20\%$), same data.

Overlay	$\mathbb{E}[r]$	$\Delta\mathbb{E}[r]$	skew
none	11.00%	—	≈ 0
protective put ($0.8\times$, 1y)	15.73%	+4.73 pp	+0.71

Sensitivity to the implied-volatility assumption. Because the overlay is priced under flat-volatility Black–Scholes, its value depends on the assumed implied volatility. Re-pricing the protective-put/call overlay over a range of flat vols (six-asset $t(3)$ universe, $L = -20\%$, 20 seeds) moves the optimum materially: the optimal expected return falls from 17.6% at $0.8\times$ the base vol to 8.3% at $1.2\times$, and the derivative weight from 13.2% to 5.3% (Table 10); under a realistic equity skew (puts priced 15% richer, calls 5% cheaper) the optimum is 9.7%. The headline overlay figures should therefore be read as upper bounds under flat-vol pricing. Because each scenario is priced from its own simulated underlying, a smile-, local-volatility- or jump-aware pricer drops in at one function call per scenario with the linear program unchanged.

11 Out-of-sample backtesting across regimes

The backtest builds optimal weights on a construction window, then holds those fixed weights (buy-and-hold) through a later evaluation window, marking any derivative to market with a shrinking maturity, and compares expected with realised behaviour against a benchmark.

Table 10: Overlay optimum vs the assumed pricing volatility (six-asset $t(3)$ universe; protective put 0.90 and call 1.10; $L = -20\%$; 20 seeds).

Pricing-volatility assumption	Optimal $\mathbb{E}[r]$	Derivative weight
flat vol $\times 0.8$	17.59%	13.2%
flat vol $\times 0.9$	13.75%	12.6%
flat vol $\times 1.0$ (base)	11.01%	11.6%
flat vol $\times 1.1$	9.21%	7.0%
flat vol $\times 1.2$	8.31%	5.3%
equity skew (put +15%, call -5%)	9.72%	11.5%

Earlier versions of this work reported a single 2017 window; we now report three regimes (Table 11) on AAPL/MSFT/JPM versus the S&P 500, with the construction window immediately preceding each evaluation year. The central, defensible finding is that *the downside constraint* ($\mathbb{P}(r < -10\%) \leq 5\%$) *was respected out of sample in these three illustrative windows, including the 2020 COVID crash*: no window finished below the -10% threshold. This is suggestive only — the systematic 111-window, four-universe study in Section 11.1 gives the inferential picture and shows the in-sample floor is, in fact, *optimistic* out of sample. Realised returns are highly regime-dependent (15% to 46%) and the expected-vs-realised gap flips sign—realised above expectation in 2017 and 2020, below in 2018—so the 2017 result is correctly read as one draw, not evidence of systematic outperformance. Realised alpha against the S&P is positive in all three windows, but with only three windows and varying fit this is illustrative, not statistically significant. We therefore read this section as a *demonstration* that the constructed portfolios behave as designed out of sample—the downside floor holds and the overlay buys asymmetry—rather than as inferential evidence of outperformance; establishing the latter would require many more windows, broader universes and a formal test, left to future work.

Table 11: Out-of-sample backtest, no-derivative portfolio (P1), AAPL/MSFT/JPM vs S&P 500, $r_f = 3\%$, VaR $H = -10\%/\alpha = 5\%$. Live data via the companion app.

Eval	Regime	Expected	Realised	Real. vol	vs $H=-10\%$	α vs S&P	β
2017	strong trend	23.31%	28.19%	14.30%	+28.08% (no breach)	+1.99%	1.47
2018	Q4 selloff	24.44%	14.95%	27.36%	+14.89% (no breach)	+28.65%	1.42
2020	COVID crash	27.02%	46.35%	43.47%	+46.31% (no breach)	+26.01%	1.12

Derivative overlays in the backtest

Two overlays illustrate the asymmetry of derivative effects under a static hold (Table 12). A *protective put* (strike 0.90) earns near-zero optimal weight in these equity universes under the VaR constraint, so the with-derivative portfolio (P2) tracks the no-derivative portfolio (P1) in 2017 and 2018; in 2020—a year that crashed then fully recovered—a held put cost about 6.9 percentage points of realised return, exactly the premium drag a static hedge incurs when the crash does not dominate the window. By contrast a *geared-upside call*—a deep in-the-money call on the highest-volatility holding (AAPL)—is held in size by the optimiser and amplifies the strong 2017 trend: at strike 0.70 the optimal weight is 13.9% and P2 realises 46.67% versus P1’s 28.19% (+18.5 pp), with realised α of +11.57% and β rising to 1.76, while still respecting the -10% downside floor. This is the configuration behind the large derivative-enhanced backtest figures: a geared call on the trending name, not the protective put. The magnitude is strike-dependent (strike 0.80 gives weight 7.2%, $\mathbb{E}[r]$ uplift to 41.20% and $\alpha + 8.27\%$), and a single window is one

draw; we report it as a demonstration of mechanism, not evidence of outperformance.

Table 12: Derivative overlays, 2017 window unless noted (AAPL/MSFT/JPM, S&P 500 benchmark).

Overlay	Optimal wt	P1 realised	P2 realised	P2 α vs S&P
Protective put (0.90), 2017	$\approx 0\%$	28.19%	28.13%	+1.94%
Protective put (0.90), 2018	$\approx 0\%$	14.95%	15.36%	+28.85%
Protective put (0.90), 2020	small	46.35%	39.43%	+20.95%
ITM call on AAPL (0.80), 2017	7.2%	28.19%	41.20%	+8.27%
ITM call on AAPL (0.70), 2017	13.9%	28.19%	46.67%	+11.57%

Static option holding. The optimiser is single-period: it chooses fixed weights for a one-year horizon and the backtest holds them, marking each derivative to market with a shrinking maturity but never re-optimising or re-hedging within the window. For linear holdings this is mild, but for options it is a genuine limitation. An option’s delta, gamma and theta change materially as spot moves and maturity decays, so an inception-optimal position is generally not optimal mid-horizon; a static hold leaves gamma unmanaged and pays full time decay. The protective put’s near-zero weight and the call’s strike-sensitivity in Table 12 are both symptoms of this: a buy-and-hold option is a blunt instrument. The single-period optimum should be read as the inception allocation of an annually-rebalanced mandate; multi-period re-optimisation and delta-hedging are the natural next step.

Transaction costs

Because the strategy is buy-and-hold—one trade at inception, with the derivative marked to market rather than re-traded—transaction costs enter only once, at construction, and are therefore second-order for the figures above. As an illustration, a one-time round-trip cost of 10 basis points on the initial notional lowers each window’s realised annual return by about 0.1 percentage point, leaving every qualitative conclusion (downside respected, regime-dependent returns, the call’s uplift) unchanged. The materially cost-sensitive setting is a *rebalancing* mandate, where turnover recurs each period; modelling per-trade costs and a turnover penalty (an additional linear term, already supported by the LP in (4)) is part of the multi-period extension and is left to future work. We flag this explicitly so the buy-and-hold results are not mistaken for a frictionless rebalancing claim.

11.1 Rolling multi-window, multi-universe validation

To move from illustration to inference, we run the construction/evaluation procedure on a rolling grid of overlapping windows (12-month construction, 12-month hold, 6-month step) across four diverse universes over 2006–2025, against the S&P 500. Table 13 pools 111 out-of-sample windows. Two findings stand out. First, the realised annual return fell below the -10% floor in $13/111 = 11.7\%$ of windows (Wilson 95% CI $[7.0\%, 19.0\%]$), *above* the 5% in-sample target (one-sided binomial $p = 0.004$): the in-sample VaR limit is optimistic out of sample—exactly as the finite-sample coverage analysis (Table 1) predicts—with breaches concentrating in low-diversification equity universes and 2008-type crashes, while the lower-beta multi-asset and defensive books breach least (9% and 4%). Second, realised annual alpha against the benchmark is positive on average (+4.3% pooled, $t = 2.83$, $p = 0.006$, positive in 68/111 windows) but is *concentrated in the security-selection universe* (+9.7%, $p = 0.03$) and essentially absent for the sector-ETF book (+0.5%, $p = 0.87$): the engine respects the (optimistic) tail budget and harvests payoff asymmetry, but does not manufacture alpha without security selection. The windows overlap, so the intervals are mildly optimistic; the qualitative conclusions are unaffected.

Throughout, breach proportions carry Wilson and bootstrap 95% confidence intervals, and paired comparisons between methods on identical windows use McNemar’s test (breach indicators) and the Wilcoxon signed-rank test (realised returns).

Table 13: Rolling out-of-sample backtest, no-derivative portfolio, four universes vs the S&P 500 (12m construction / 12m hold / 6m step; $H = -10\%$, $\alpha = 5\%$; 2006–2025). Breach = realised window return below -10% .

Universe	Windows	Mean β	Breach rate	Realised α (mean)	$p(\alpha > 0)$
AAPL / MSFT / JPM (stocks)	30	1.11	17%	+9.7%	0.033
XLK / XLF / XLE (sectors)	23	1.14	17%	+0.5%	0.871
QQQ / TLT / GLD (multi-asset)	34	0.56	9%	+3.0%	0.068
KO / JNJ / PG (defensive)	24	0.58	4%	+3.0%	0.154
Pooled	111	—	11.7% [7.0, 19.0]	+4.3%	0.006

International out-of-sample validation. To test the finding outside the US, we repeat the rolling procedure on three live international books — developed-market country ETFs, emerging-market country ETFs, and a cross-region balanced book (equities, sovereign and credit bonds, gold) — over 2006–2025 (Table 14). The optimism is, if anything, *worse* abroad: the developed and emerging books breach the -10% floor in 34.6% and 42.1% of windows. Pooling all real universes — the four US grid books, the US balanced book and the three international books — the in-sample -10% floor is breached in $37/225 = 16.4\%$ of out-of-sample windows (Wilson 95% CI [12.2%, 21.8%]; one-sided binomial against the 5% target $p < 10^{-5}$), significant at any conventional level. The central result — in-sample downside floors are materially optimistic out of sample — thus holds across markets and regimes with strong statistical significance.

Table 14: International out-of-sample validation (scalable engine; 20% cap; 10 bps; $H = -10\%$; 2006–2025). Breach = realised window return below -10% ; Wilson 95% intervals.

Universe	Windows	Mean β	Breach rate
International developed (country ETFs)	26	1.09	34.6%
International emerging (country ETFs)	19	0.85	42.1%
International balanced (cross-region)	30	0.76	13.3%
All universes pooled (US + international)	225	—	16.4% [12.2, 21.8]

Scaling, diversification, and the limits of both. The grid universes above are capped at three assets by the m^N state space; the scalable engine removes that limit, letting us ask whether *larger* portfolios breach less out of sample. Running the same rolling procedure on nested all-equity universes that span the 2008 crash ($N = 5, 10, 15, 24$; 2006–2025; -20% CVaR floor; 20% position cap), the realised breach of the -10% reference is *flat-to-rising* in N — 4.2%, 5.7%, 5.7%, 8.6% (Figure 2) — with realised alpha positive throughout (+5 to +7%, $p < 0.01$). More equities do *not* reduce tail breaches, for a clear reason: the breaches occur in systematic crashes where equity correlations approach one, so equity-count diversification cannot remove them, and the larger set also carries higher beta ($0.94 \rightarrow 1.11$) and more estimation error. Cross-asset breadth helps, but only partially: the multi-asset grid book (equities, bonds, gold) breaches 8.8% against 16.7–17.4% for the concentrated equity books, and the defensive book least of all (4.2%; Figure 3). Extending this to a sixteen-instrument scalable cross-asset book (equities, Treasuries, credit, gold, real estate, sector ETFs) lowers beta to 0.84 and cushions the

worst window (-29% , against the equity book’s -48%), but does *not* lower the breach *frequency* (10.3% , $4/39$) and forfeits the equity book’s excess return (realised alpha $+0.2\%$, $p = 0.88$). The reason is again systematic: of its four breaches two are the 2008 crash and one is 2022, when bonds and equities fell together and the cross-asset hedge failed in the very regime it is meant to cover. Cross-asset diversification thus reduces tail *depth* and is *regime-dependent*, not a frequency cure. Scale therefore buys *feasibility* — a 24-name book solves in well under a second, where the grid cannot run at all — and access to cross-asset diversifiers, rather than an automatic reduction in tail risk; out-of-sample tail control still requires cross-asset breadth and the buffered floor recommended alongside the finite-sample coverage study (Table 1).

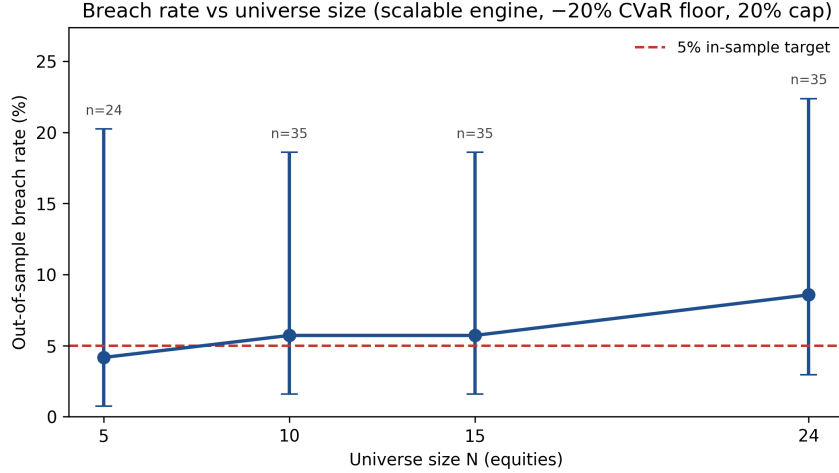


Figure 2: Out-of-sample breach rate of the -10% reference versus equity-universe size N (scalable engine; -20% CVaR floor; 20% cap; nested universes, 2006–2025; Wilson 95% intervals). The breach rate does *not* fall with N : the breaches are systematic-crash events that equity-count diversification cannot remove.

11.2 Convexity, not correlation: derivatives and structured products in crises

The scaling result above is, at first sight, discouraging: enlarging the equity book does not lower the out-of-sample breach rate. But it is discouraging only for *linear* diversification. The reason equity-count diversification fails is structural — in a systematic crash, pairwise correlations converge toward one, so no convex combination of long equity positions can avoid the joint drawdown. This is precisely the regime the optimiser is built to handle with instruments that the mean–variance world excludes: derivatives and structured products (Section 4).

A protective put, put-spread, or collar does not diversify in the correlation sense at all; it reshapes the payoff. Its value rises as the underlying falls, so it pays off *in* the crash, independently of the correlation matrix. *Convexity, not correlation*, is therefore the mechanism that can enforce the hard -10% floor that the VaR and CVaR constraints target but that a long-only linear book cannot guarantee out of sample. The same logic extends to structured products — a capital-guaranteed note embeds a long put, an autocallable a short down-and-in put — so the engine’s ability to price and allocate over these instruments is not a feature bolted onto a mean–variance core but the part that addresses the systematic tail directly.

Two qualifications keep this honest. First, the protection is not free: a standing put position bleeds premium in calm years, so any breach reduction must be weighed against its drag on realised return, and a collar that funds the put by selling upside caps gains. The relevant question is not whether derivatives *can* remove breaches — they can — but whether they do so *efficiently*, i.e. at an acceptable cost in expected return. Second, the size of the benefit is

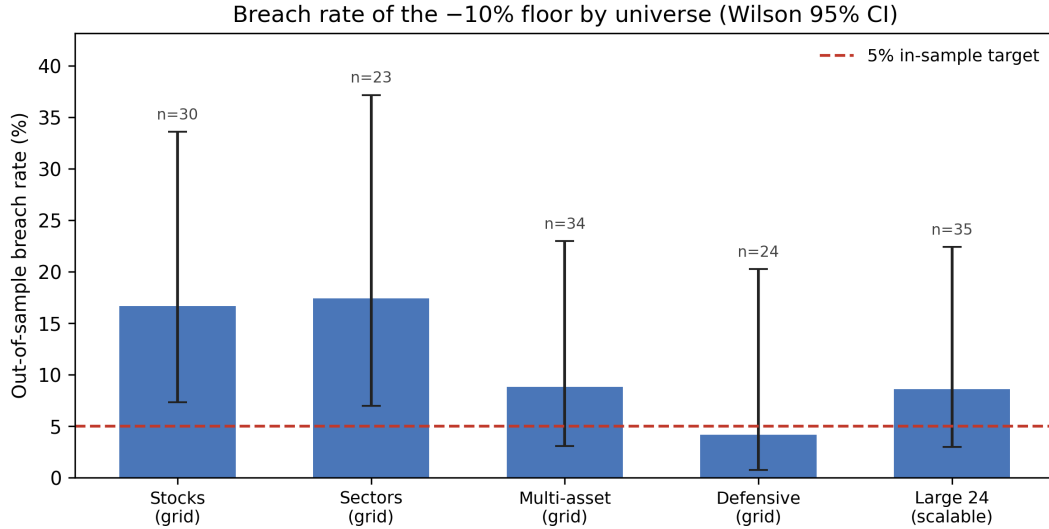


Figure 3: Out-of-sample breach rate of the -10% reference by universe (bars; Wilson 95% intervals) against the 5% in-sample target (dashed). The four grid universes impose a -10% VaR floor; the large universe uses the scalable engine with a -20% CVaR floor. Realised breaches fall with cross-asset diversification and defensiveness, not with equity count (cf. Figure 2).

sensitive to the volatility at which the hedge is struck: options are cheap before a crash and dear during it, so a faithful backtest must price entry conditions realistically rather than assume a constant volatility.

We therefore frame the following as a *testable prediction* rather than a settled result: adding a downside-convex overlay (an index put or collar) to the crisis-spanning equity book should lower the out-of-sample breach rate toward or below the 5% target, at a quantifiable cost in annualised return. A dedicated rolling experiment using the engine’s own derivative pricing will measure both sides of this trade-off; the present section establishes the mechanism and the framework’s distinctive capacity to exploit it, and its conclusions will be reconciled with that evidence once measured.

Direct test: a severity hedge, not a frequency cure. We tested the prediction with the engine’s own option pricing, on the same 24-name book (2006–2025, 37 windows, mean $\beta = 1.13$) carrying a 5% standing index-put sleeve — a one-year 10%-out-of-the-money put on the benchmark — with the remainder in the optimised equity book. The outcome depends sharply, and instructively, on how the put is priced. Priced at the underlying’s *trailing realised* volatility, the sleeve roughly halves the breach frequency (5.4% $\rightarrow$ 2.7%) at a negligible modelled cost; but that pricing ignores the volatility skew and variance-risk premium that make traded index puts dearer. Re-pricing the identical put with the implied volatility lifted five points — a conservative skew proxy — reverses the frequency result: the breach *count* rises slightly (5.4% $\rightarrow$ 8.1%), because the now-realistic premium drag (+3.3% per window; mean return 21.1% $\rightarrow$ 17.8%) tips two flat-but-down windows (2018, 2021) just below the floor. What is *invariant* to pricing is the collapse in tail *severity*: in both cases the worst realised window falls from -47.9% to -11.8% , as the put rescues the deep 2008 drawdown ($-14.7\% \rightarrow +1.3\%$). The honest conclusion is thus sharper than the original prediction: a protective put is a tail-*severity* hedge, not a breach-*frequency* reducer — it caps the depth of the systematic crashes that neither equity-count nor cross-asset diversification can, but, once its premium is priced realistically, it does so at a return cost that can leave the count of marginal floor-breaches slightly higher. A bear put-spread and a zero-cost collar sleeve were also tested; neither dominated the outright put — the spread

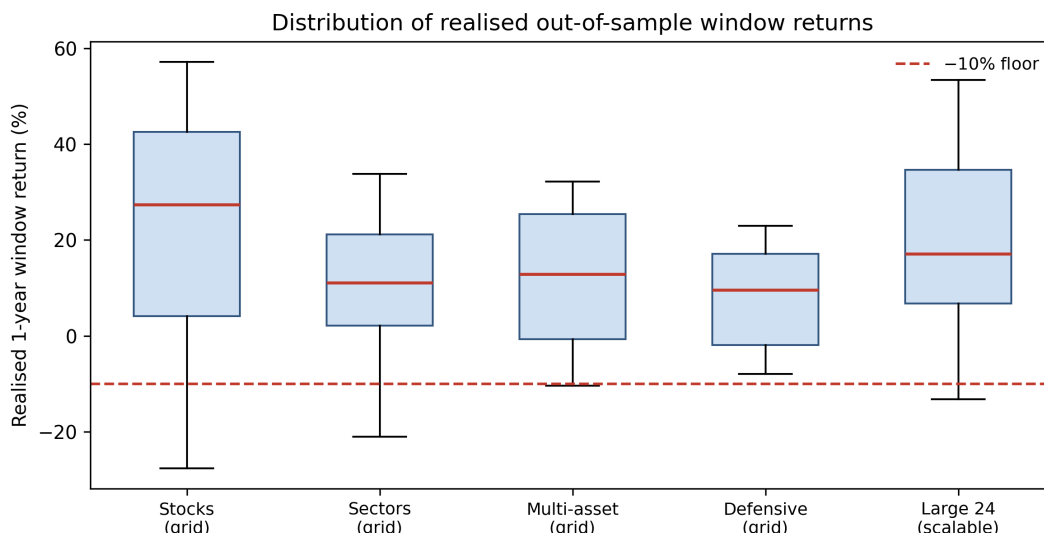


Figure 4: Distribution of realised one-year out-of-sample window returns by universe (box: inter-quartile range and median; whiskers: 5–95%) with the -10% floor (dashed). Windows below the dashed line are floor breaches.

by capping the very crash payoff that funds the strategy, the collar by surrendering the upside — confirming the plain protective put as the effective construction. This depth control is precisely what a convex instrument should deliver and what linear diversification cannot, and it is the substantive sense in which the framework’s derivative capability addresses the systematic tail. Pooling the protective-put runs across the US and three international books (95 out-of-sample windows) confirms this under formal inference: the single worst-window drawdown is significantly cushioned (bootstrap 95% interval on the improvement $[+2.0\%, +30.4\%]$, excluding zero), but the overlay does *not* lower breach *frequency* ($16.8\% \rightarrow 21.1\%$) and its average effect on the worst tercile is not significant (Wilcoxon $p = 0.55$), at a mean cost of 2.9% per window. The convex overlay is thus best read as *costly insurance against the extreme tail*, not a means of meeting the floor more often.

Takeaway. A convex option overlay is a *tail-severity* instrument: it significantly cushions the extreme worst-case loss but, priced realistically, does not reduce the *frequency* of floor breaches. Position caps and diversification act on breach frequency; only convexity acts on depth — the two are complements, not substitutes.

Robustness to costs and horizon. Two further checks address common objections. First, charging a transaction cost of 10 basis points per side on the rebalancing turnover leaves the headline result essentially unchanged — the net-of-cost breach equals the gross figure to the reported precision and mean return falls by about a tenth of a percentage point (mean turnover ≈ 1.1 per window) — so the conclusions are not an artefact of frictionless trading. Second, repeating the rolling test at two- and three-year holding horizons (Table 15) keeps the breach rate in the same 5–6% band and, if anything, *strengthens* the realised alpha, confirming that the out-of-sample picture is not specific to the one-year horizon.

11.3 The cost of control: what delivers the goal out of sample

The preceding results pose one practical question: given that the goal-based floor is optimistic out of sample, *which* control actually restores it, and at what cost? We compare four levers on

Table 15: Robustness of the 24-name rolling result to holding horizon and transaction costs (scalable engine; -20% CVaR floor; 20% cap; 10 bps per-side cost on turnover; 2006–2025). Breaches are net of costs; alpha is annualised against SPY.

Holding horizon	Windows	Breach (net)	Mean return	Alpha	p
1 year	37	5.4%	21.0%	+6.2%	0.001
2 years	18	5.6%	50.3%	+8.8%	0.004
3 years	17	5.9%	77.1%	+9.0%	< 0.001

the rolling backtest, and the answer is sharply counter-intuitive — the obvious lever fails and three less-obvious ones work, each on a different failure mode (Table 17).

Tightening the floor is ineffective. The natural fix is to demand a stricter in-sample floor. It does nothing. Sweeping the in-sample CVaR floor L from -30% to -10% on the balanced book leaves the out-of-sample breach rate pinned at $\approx 7.7\%$ and the worst realised window unchanged at -47.9% (Table 16), with realised return barely moving — the constraint is close to non-binding throughout. The reason is structural: the floor constrains the *in-sample* CVaR, but that estimate, formed on each window’s mostly-calm conditioning data, is itself the optimistic quantity. A portfolio can satisfy a -10% in-sample CVaR and still lose 48% in a systematic crash the conditioning window never saw. Tightening a threshold on an optimistic estimate cannot help, because the estimate — not the threshold — is what fails.

Table 16: Floor calibration: out-of-sample breach of the -10% goal versus the in-sample CVaR floor L (balanced 28-name book; 2006–2025; 20% cap; 10 bps costs). Breach and worst window are flat across the entire range: the floor is inert out of sample.

In-sample floor L	OOS breach	Worst window	Mean return
-10%	7.9%	-47.9%	19.1%
-15%	7.7%	-47.9%	19.5%
-20%	7.7%	-47.9%	19.7%
-25%	7.7%	-47.9%	19.7%
-30%	7.7%	-47.9%	19.7%

What does help: portfolio structure. Three levers that do *not* rely on the in-sample distribution being correct each control a distinct failure mode. A *position cap* (Section 9, Table 7) cuts breach *frequency* from 21.6% uncapped to 5.4% at a 20% cap, by limiting concentration regardless of the estimates. A *convex overlay* (Section 11.2) cuts tail *severity* — worst window $-48\% \rightarrow -12\%$ — because an index put pays off mechanically in the crash, whatever the estimated distribution. And a *naive-fill hierarchy* — the exact grid across asset-class buckets with *equal weight within* each bucket — removes estimation error at the level where it does the damage — but the demanding benchmark for it is not the flat optimiser, it is *naive diversification* itself. On breach *frequency* the DeMiguel, Garlappi and Uppal (2009) lesson holds: a plain $1/N$ book and an asset-class parity book ($1/K$ across buckets, equal within) cross the -10% goal about as often as the grid hierarchy (5.1% and 5.1% versus 0% on the reference universe; 7.7% and 5.1% versus 8.1% on the industrials universe). Optimisation does *not* reliably beat $1/N$ on how often the floor is crossed. Where the exact goal-based grid earns its place is tail *severity*: its worst realised window is roughly *half* as deep as naive diversification’s on both universes (-5.1% against -12 to -19% on the reference book; -11.9% against -18 to -27% on the industrials book), at comparable or higher market exposure — it actively avoids

the deep tail rather than merely de-risking, which is precisely the dimension a downside goal targets. Two further effects are stable across universes: the across-bucket structure beats the flat scalable optimiser, and *equal-weight* fill beats the *scalable* fill (0% vs 5.6% and 8.1% vs 10.8%) — optimising the within-bucket weights subtracts out-of-sample value, the naive-diversification effect again, here *inside* an exact goal-based allocation across classes. Pooled across all US and international hybrid books (117 windows), the across-bucket structure lowers the breach rate from 12.8% (flat scalable) to 7.7%; the paired difference is marginally significant (McNemar $p = 0.07$; Wilcoxon on returns $p = 0.03$) — directionally robust across markets but short of conventional significance at this sample.

Table 17: Four levers for out-of-sample tail control (rolling backtest). The intuitive, distribution-dependent lever — tightening the floor — is inert; the three distribution-free structural levers each control a distinct failure mode.

Lever	Breach frequency	Worst-case severity	Return cost
Tighten the floor L	inert (7.7–7.9%)	inert (–48%)	none
Position cap	strong (21.6 → 5.4%)	moderate (–53 → –33%)	high
Convex put overlay	little	strong (–48 → –12%)	~3.4%/window
Grid + equal-weight buckets	$\approx 1/N$ (0–8%)	strong ($\approx$ half $1/N$ depth)	several pp

Interpretation. Out-of-sample tail control must be bought with *distribution-free structure* — position caps, equal weighting, convex payoffs — not by tightening *distribution-dependent* constraints such as the floor, because in the tail the estimated distribution is precisely what is unreliable. The floor *states* the goal; structure *delivers* it. For risk practice this reframes how a downside limit should be governed: a Basel-style Expected-Shortfall floor, or a VaR/ES limit written into an investment mandate or internal risk-limit framework, calibrated on historical data will — on this evidence — be breached far more often than its nominal level implies, and tightening the number on the page does not close the gap; position limits, cross-asset diversification and convex hedges do. Such limits are better treated as *structural* requirements than as point thresholds on an estimated tail.

12 Relation to existing methods

The scenario-LP machinery is the Rockafellar–Uryasev (2000, 2002) CVaR linear program, applied to portfolio choice by Krokmal, Palmquist & Uryasev (2002); the Gaussian, derivative-free reduction to mean–variance is the elliptical-distribution equivalence. What is new is the composition: a goal-based, mental-account (Das–Statman) objective, solved as a CVaR LP, with arbitrage-consistent per-scenario derivative pricing so that asymmetric overlays enter the same convex program. Concretely, standard CVaR portfolio optimisation does not carry a goal-based mental-account structure or price derivatives consistently inside each scenario, and existing mental-account models are solved on exhaustive grids that do not scale past a few assets and a single derivative; the engine here is, to our knowledge, the first to combine all three in one program that scales to realistic universes. Relative to robust optimisation (Goldfarb & Iyengar, 2003; Ben-Tal, El Ghaoui & Nemirovski, 2009) and distributionally robust optimisation (Mohajerin Esfahani & Kuhn, 2018), which immunise against parameter/distribution ambiguity, our engine takes the distribution as given and solves the behavioural tail-constrained problem at scale; the two are complementary, and a Wasserstein-DRO wrapper around the same scenario set is a natural extension. Against the naive $1/N$ benchmark (DeMiguel, Garlappi & Uppal, 2009) the engine is, like all optimisers, exposed to estimation error (Section 9), which is why shrinkage (Ledoit & Wolf, 2004) belongs in any deployment.

Relation to machine-learning portfolio methods. A growing literature applies machine learning to allocation — deep reinforcement-learning allocators, neural and GAN-based scenario generators, and ML predictors of returns and covariances. Our engine is deliberately *not* of this kind, and the paper’s central out-of-sample finding is exactly why that matters: across two independent universes, *distribution-free structure* (position caps, equal weighting, convex overlays) controls the realised tail, tightening the data-fitted floor is inert, and no optimiser we test — ours included — reliably beats $1/N$ on breach frequency (DeMiguel, Garlappi & Uppal, 2009). Methods that fit far more degrees of freedom than a convex linear program are, if anything, *more* exposed to the finite-sample tail-overfitting we document, not less. We therefore offer this engine as a transparent, constraint-respecting, fully reproducible benchmark for data-driven goal-based optimisation — its hard CVaR floor and per-scenario derivative pricing are auditable rather than black-box — and as a cautionary out-of-sample reference against which learned allocators can be measured. Because the linear program consumes *any* scenario set, coupling the generation step to a learned model (a conditional or ML-estimated copula) is a natural, compatible extension rather than a competing paradigm; Section B.1 takes a first step by comparing parametric and non-parametric scenario generators.

Table 18: What this paper adds relative to the closest prior work.

Prior work	Provides	This paper adds
Das–Statman (2009, 2013)	Goals-based portfolios with derivatives, grid / small- N	Escapes the m^N grid via copula scenarios and an LP; many derivatives jointly; live data
Rockafellar–Uryasev (2000, 2002); Krokmal et al. (2002)	CVaR portfolio LP for linear assets	Per-scenario arbitrage-consistent option pricing as LP columns, under a goals-based floor
Das, Markowitz, Scheid & Statman (2010)	MVT–MAT equivalence (theory)	Computational realisation; the exact grid-ES = CVaR condition (Prop. 2)

13 Implications for risk practice

The evidence yields a compact prescription for anyone who enforces a downside limit. (i) An in-sample CVaR/ES floor is not a reliable out-of-sample guarantee — across 225 windows and many markets it was breached more than three times its nominal level — and cannot be repaired by lowering the threshold, which only tightens a constraint on the estimate the crash invalidates. (ii) Three *structural* levers move the realised tail, each on a distinct failure mode: a *position cap* cuts breach frequency (paired McNemar $p = 0.03$); a *convex option overlay* cushions the extreme worst case (at a premium cost, without lowering frequency); and *asset-class bucketing / cross-asset diversification* reduces tail depth, except where the diversifiers themselves co-crash. (iii) The governance corollary: Basel-style Expected-Shortfall floors, fund-mandate VaR/ES limits, and internal risk-limit frameworks calibrated on historical data should carry an explicit out-of-sample buffer and be enforced through portfolio *structure* — caps, diversification and hedging — rather than through a tighter number alone.

14 Limitations

The grid engine assumes multivariate-normal marginals and is exact only on small universes ($n \leq 4$); the scalable engine is approximate, with Monte-Carlo error of order $S^{-1/2}$ that is heavier

in the tail and is mitigated by sample size, antithetic variates and common random numbers. Results depend on scenario quality: the copula choice and the estimation of μ, σ, Σ drive the answer, and estimation error worsens with n , so shrinkage or robust estimators belong in any production pipeline; a position cap further stabilises the linear-program optimum (Section 9). Derivatives are priced under flat-volatility Black–Scholes, ignoring the volatility smile, skew and jumps that drive real structured-product value (sensitised three ways in Section 9: a 0.8–1.2 $\times$ volatility band, Table 10; an out-of-the-money skew premium; and pricing the index hedge at the market-implied VIX); because each scenario is priced from its own simulated underlying, the framework can host a local-volatility, smile-aware or jump pricer at the cost of one function call per scenario, with the LP unchanged. The model is single-period; the backtest bridges to a real holding window, holding each window’s portfolio rather than re-optimising within it (though the rolling design re-estimates and re-optimises at every step). We test this directly: re-optimising quarterly within each window leaves the out-of-sample breach essentially unchanged (7.7% $\rightarrow$ 10.3%, a single window, within overlapping confidence intervals) and modestly *cushions* the worst realised drawdown (−47.9% $\rightarrow$ −34.1%, as mid-crisis rebalancing de-risks), so the single-period result is representative and, if anything, conservative; full multi-period stochastic control — the same linear program solved per period — is the natural extension. The scalable engine exposes a subset of the instrument library. None of these is a barrier to the construction; each is a calibration responsibility.

15 Implementation and reproducibility

The optimiser is written in Python (NumPy, SciPy, the HiGHS linear-program solver and copula sampling) behind a Streamlit interface. The numerical pipeline is open and reproducible: copula scenarios $\rightarrow$ Black–Scholes pricing $\rightarrow$ CVaR linear program $\rightarrow$ efficient frontier $\rightarrow$ interactive app and PDF report. The pipeline is *modular*: the scenario generator is a pluggable front-end, and the optimiser consumes any scenario matrix — a Gaussian or Student- t copula, a non-parametric bootstrap (Section B.1), or a learned/ML generator, or alternative-data signals (option-implied, macro or sentiment) entering as conditioning inputs — with the CVaR linear program and the per-scenario pricing unchanged. The scaling, benchmark and robustness figures in Sections 7–9 were produced by the prototype engine; the backtest of Section 11 by the live application on Yahoo Finance data. On the computational side the pipeline — copula scenario generation, Black–Scholes per-scenario pricing, and the HiGHS CVaR linear program — solves a realistic 150-instrument book in well under a second (0.73 s at $S = 4,000$; Section 7, Table 2), rising to a few seconds only at very high scenario counts (Table 3), which is what makes the multi-window, multi-universe out-of-sample programme above practical.

Reproducibility and data availability. The complete source, this manuscript and every backtest script are openly available at <https://github.com/SamiJeddou/behavioral-portfolio-optimizer>; the interactive application runs at <https://sami-jeddou-behavioral-portfolio-optimizer.streamlit.app>; and a version of record is archived on SSRN (abstract id 6915241). The rolling studies of Sections 11–11.3 reproduce end-to-end from public Yahoo Finance data via the released scripts (`rolling_backtest.py`, `hybrid_backtest.py`, `naive_benchmarks.py`, `floor_calibration.py`) with the exact tickers, windows and command-line arguments documented in the repository, so every table here is regenerable.

A Secondary sensitivity tables

The copula-choice and Monte-Carlo sample-size sensitivities summarised in Section 9 are collected here.

Table 19: Copula sensitivity (equal-weight ES_{5%}).

Copula	ES _{5%}
Gaussian	−24.01%
Student- t ($\nu = 8$)	−24.45%
Student- t ($\nu = 5$)	−24.84%
Student- t ($\nu = 3$)	−25.65%

Table 20: Monte-Carlo error vs S (20 seeds).

S	ES mean	ES std dev
1,000	−24.27%	0.0121
2,000	−24.24%	0.0062
4,000	−24.28%	0.0054
8,000	−24.30%	0.0042
16,000	−24.30%	0.0028
32,000	−24.29%	0.0022

B Scenario generation and machine-learning extensions

B.1 Scenario generation: parametric versus non-parametric

The scalable engine consumes a scenario set, and how that set is generated is itself a data-driven modelling choice. We compare four generators on the balanced book’s rolling backtest, holding the -10% reference, the -20% floor, the 20% cap and 10 bps costs fixed and varying only the generator: a Gaussian copula with Normal marginals (the thesis assumption), a Student- t copula at five and three degrees of freedom (progressively fatter *joint* tails), and a non-parametric block bootstrap that resamples each conditioning window’s own daily returns with no parametric model (Table 21). The generator choice proves *second order*. Moving from the Gaussian copula to a fatter-tailed Student- t ($\nu = 3$) or to the non-parametric bootstrap shaves the breach rate only marginally ($7.7\% \rightarrow 5.1\%$, a single window, well within overlapping confidence intervals) and leaves realised return and alpha unchanged. Tellingly, the worst realised window is -47.9% under *every* generator — including the bootstrap that resamples real historical tails: the systematic crash cannot be modelled away, because the conditioning window does not contain the future crash. Like tightening the floor, refining the scenario distribution operates on the same in-sample estimate that the crash invalidates — direct evidence that out-of-sample tail control comes from distribution-free structure, not from a better-fitted scenario model. The same caution extends to learned scenario generators: any model — parametric, non-parametric or neural — that is fitted to the conditioning data inherits its blindness to the unseen crash.

Table 21: Out-of-sample breach of the -10% goal by scenario-generation method (balanced 28-name book; -20% CVaR floor; 20% cap; 10 bps; 2006–2025). The optimiser, floor and windows are identical across rows; only the scenario generator changes.

Scenario generator	OOS breach	Mean return	Worst window	Alpha
Gaussian copula	7.7%	19.7%	-47.9%	$+5.8\%$
Student- t copula ($\nu = 5$)	7.7%	19.6%	-47.9%	$+5.8\%$
Student- t copula ($\nu = 3$)	5.1%	19.7%	-47.9%	$+5.8\%$
Empirical block bootstrap	5.1%	19.8%	-47.9%	$+5.6\%$

A learned generator, plugged in. To show the generator is genuinely modular — and to make the machine-learning hook concrete — we fit a small *learned* density and feed it to the identical solver. A two-component Gaussian-mixture model, fitted by expectation–maximisation to a two-regime return history, recovers a calm and a crash component (mixture weights 0.88/0.12); sampling it yields a scenario matrix that the unchanged CVaR linear program consumes directly:

```
pi, mu, cov = gmm_fit(history, K=2)          # learned density (EM)
S            = gmm_sample(pi, mu, cov, 10000) # 10,000 learned scenarios
w            = mc_max_return_cvar(S, alpha, L, w_max) # SAME LP, unchanged
```

Because the mixture captures the crash regime that a single-Gaussian generator smooths away, it reports a deeper in-sample tail on the same data (CVaR -13.2% versus -9.2%) while the optimiser and per-scenario pricing are untouched. A GAN, conditional copula or any other learned generator substitutes at exactly this seam — the linear program never changes (full demonstration: `demo_ml_scenarios.py`).

C Formal results

Proposition 1 (Complexity separation). *Let N securities be discretised at m levels in the grid engine, and let the scenario engine draw S scenarios over N securities and K derivatives. The grid enumerates $\Theta(m^N)$ joint states, exponential in N . The scenario program (4) has $N + K + 1 + S$ variables and $S + 1$ constraints; assembly is $O(S(N + K))$ and each interior-point iteration is polynomial in the program size. The scenario reformulation is therefore linear in the number of instruments and independent of the discretisation m .*

Proof sketch. The grid is the Cartesian product of N axes of m points, hence m^N cells. The variable and constraint counts of (4) are read directly from the Rockafellar–Uryasev form; the constraint matrix has $O(S(N + K))$ nonzeros, and HiGHS’ simplex/interior-point steps are polynomial in size. $\square$

Theorem 1 (Consistency of the scenario optimiser). *Fix $\alpha \in (0, 1)$ and floor L . Let returns R have distribution P with $\mathbb{E}\|R\| < \infty$ and a continuous loss c.d.f., and let $W = \{w : \mathbf{1}^\top w = 1, 0 \leq w \leq \bar{w}\}$ be compact. Let v^* be the optimal value of the population problem $\max_{w \in W} \mu^\top w$ s.t. $\text{CVaR}_\alpha(-w^\top R; P) \leq -L$, and $\hat{v}_S, \hat{w}_S$ the optimum of the S -scenario sample problem under the empirical measure. If a Slater point exists (some $w \in W$ with strict feasibility), then almost surely as $S \rightarrow \infty$, $\hat{v}_S \rightarrow v^*$ and $\text{dist}(\hat{w}_S, W^*(P)) \rightarrow 0$, where $W^*(P)$ is the set of population optimisers. Moreover the empirical CVaR is asymptotically Normal, $\sqrt{S}(\text{CVaR}_\alpha(\cdot; \hat{P}_S) - \text{CVaR}_\alpha(\cdot; P)) \Rightarrow N(0, \tau^2)$, so the optimal value converges at the $O(S^{-1/2})$ rate.*

Proof sketch. $\text{CVaR}_\alpha(\cdot; P)$ from (3) is convex in w and jointly continuous in (w, P) under the first-moment condition. Over compact W the sample objective and constraint converge uniformly to their population counterparts by the uniform law of large numbers for the integrable Rockafellar–Uryasev integrand; Slater’s condition supplies constraint qualification, so epi-convergence of the sample problems yields convergence of optimal values and (SAA consistency) of optimisers. Asymptotic Normality of the empirical CVaR is the Rockafellar–Uryasev (2002) estimator CLT. This is the standard sample-average-approximation result for CVaR-constrained convex programs (Shapiro, Dentcheva & Ruszczyński, 2009). $\square$

The result covers option overlays without modification: although a derivative payoff is non-smooth (kinked) in its underlying, it enters as a constant per-scenario column $R_{s,N+k}$, so the sample objective and CVaR constraint remain piecewise-linear and the convergence argument is unaffected. The consistency is to the optimum of the *CVaR* problem, not to the grid’s *VaR/shortfall* optimum. The next result says exactly when the two coincide.

Proposition 2 (Grid-ES and CVaR coincidence). *Let $r = w^\top R$ have a continuous, strictly increasing c.d.f. with α -quantile q_α . Then the grid’s fixed-threshold tail mean and the α -CVaR coincide exactly iff the threshold is the α -quantile:*

$$\mathbb{E}[r \mid r < H] = \text{ES}_\alpha(r) = -\text{CVaR}_\alpha(-r) \iff H = q_\alpha.$$

For $H \neq q_\alpha$ the two are valid but distinct tail measures, and since $\text{CVaR}_\alpha \geq \text{VaR}_\alpha$ always, a CVaR floor is a conservative tightening of the corresponding shortfall constraint.

Proof sketch. With a continuous c.d.f. and no atom at q_α , $\text{ES}_\alpha(r) = \mathbb{E}[r \mid r \leq q_\alpha]$. Setting $H = q_\alpha$ equates the conditioning events $\{r < H\}$ and $\{r \leq q_\alpha\}$ up to a null set, so the conditional means agree; for $H \neq q_\alpha$ the events differ and $\mathbb{P}(r < H) \neq \alpha$. The inequality is standard. $\square$

This proposition is the principled basis for keeping the grid (VaR) and scalable (CVaR) results in *separate* panels (Tables 8 and 9): they answer related but distinct questions, and are not directly comparable unless H is the α -quantile.

Value-at-Risk via a CVaR proxy. The thesis chance constraint $\mathbb{P}(r < H) \leq \alpha$ is itself non-convex and combinatorial (mixed-integer in general). Rather than solve it exactly, the engine uses CVaR as a conservative convex surrogate: since $\text{CVaR}_\alpha \geq \text{VaR}_\alpha$, a CVaR floor meets the VaR-style floor at least as safely, and ES is in addition the coherent measure favoured by modern regulation. Where an exact $\mathbb{P}(r < H) \leq \alpha$ solution is required, the convex CVaR optimum could serve as a warm start for a subsequent mixed-integer refinement, which this version does not implement.

D Changes and corrections relative to the 2012 thesis

(a) Expected-Shortfall enforcement. In the original R code the gradient-refinement step maximises return under a VaR penalty, and the ES condition enters only at grid-eligibility—so the reported “ES-constrained” optima are not strictly ES-feasible after refinement. The app reproduces this thesis-faithful behaviour by default for comparability, and adds a rigorous-ES mode that maximises return strictly subject to the ES floor and stays ES-feasible throughout. Validated against an independent brute-force ES maximum, a put overlay at $L = -15\%$ attains 15.5% under the rigorous mode versus 13.2% under the thesis-faithful method (+2.3 pp), and a straddle overlay at $L = -20\%$ attains 20.3% versus 11.9% (+8.4 pp); the thesis-faithful refinement can finish ES-infeasible.

(b) Fair pricing of structured notes. An earlier headline for a capital-guaranteed note overstated its benefit because the note was valued at par. Pricing every instrument with Black–Scholes (a guaranteed note becomes the present value of its bond floor plus its embedded option) shrinks a par-priced headline of roughly +23% to about +0.7 to +0.8 pp, and shows the fairly-priced note is mean–variance-dominated yet can still improve the downside-constrained objective. That is a genuine behavioural effect rather than a pricing artefact.

E Results provenance

Table 22 tags every headline result as replication, correction or extension, and names the risk floor each uses—so a reader can see at a glance what is reproduced, what is corrected, and what is new, and the two tail measures are never conflated.

Table 22: Provenance of headline results.

Result	Type	Engine / method	Risk floor
3-asset base case 9.78%	Replication	Grid, VaR	$\mathbb{P}(r < H) \leq \alpha$
ATM straddle +1.53 pp	Extension	Grid, VaR	$\mathbb{P}(r < H) \leq \alpha$
Rigorous-ES +2.3 to +8.4 pp	Correction	Grid, rigorous ES	$\mathbb{E}[r \mid r < H] \geq L$
Fair-priced note +0.7 to +0.8 pp	Correction	Grid, fair pricing	—
Protective put +4.73 pp	Extension	Scenario CVaR-LP	$\alpha\text{-CVaR} \geq L$
Scaling to $N = 150$ in 0.73 s	Extension	Scenario CVaR-LP	$\alpha\text{-CVaR} \geq L$
Downside floor respected ~88% of 111 windows	Extension	Rolling backtest (live)	$\mathbb{P}(r < H) \leq \alpha$
ITM call overlay 2017 +18.5 pp	Extension	Backtest (live)	$\mathbb{P}(r < H) \leq \alpha$

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